

Experiment No: 15

Title

Write a Python Program to Implement Decision Tree

Aim

To develop a Python program to implement a **Decision Tree Classifier** for classifying data and predicting the output.

Objective

To build and test a Decision Tree model using the Iris dataset and evaluate its prediction accuracy.

Algorithm

- Step 1:** Import the required libraries.
- Step 2:** Load the Iris dataset.
- Step 3:** Split the dataset into training and testing sets.
- Step 4:** Create a Decision Tree classifier.
- Step 5:** Train the classifier using the training data.
- Step 6:** Predict the output for the testing data.
- Step 7:** Calculate and display the accuracy.
- Step 8:** Stop the program.

Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load Dataset
iris = load_iris()
```

```
X = iris.data
y = iris.target

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)

# Create Decision Tree Model
model = DecisionTreeClassifier(random_state=42)

# Train Model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Predicted Classes:")
print(y_pred)

print("\nAccuracy:", round(accuracy * 100, 2), "%")
```

Sample Input

No user input.

(Iris dataset is loaded automatically.)

Sample Output

```
Predicted Classes:  
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0 0 0 1 0 0 2 1  
 0 0 0 2 1 1 0 0]  
Accuracy: 100.0 %  
>>>|
```

Result

Thus, the Python program to implement the **Decision Tree Classifier** was executed successfully, and the classification accuracy was obtained correctly.

